



Society of Petroleum Engineers

SPE-228867-MS

Advances in Hydraulic Fracturing Mapping: Integrating Multi-Scale Technologies for Optimal Reservoir Performance

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/228867-MS](https://doi.org/10.2118/228867-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

This paper assesses cutting-edge hydraulic fracturing mapping technologies to evaluate fracture geometry, cluster/stage efficiency, reservoir layering contribution, and frac-driven interactions (FDI). We evaluate the positive aspects, limitations, and applicability of DNA sequencing, geochemical analysis, molecular tracers, fiber optics, nano sensors, and a novel floating magnetometer sensor for optimizing injection or production, and reservoir management while mitigating environmental risks across different shale plays.

Aside from 3D proppant frac mapping, which is being developed and tested in the field scale, we could use either DNA sequencing or geochemical fingerprinting to map reservoir layering and fluid migration, with molecular tracers tracking stage connection, and/or fiber optics and/or sensors monitoring real-time FDI and fracture propagation. A unique patented floating magnetometer sensor (in development) measures stage production contribution. How these integrated technologies would compare to traditional micro-seismic and pressure-based mapping, assessing accuracy, cost, and scalability. Selection considers reservoir complexity, data resolution needs, and environmental constraints.

Field case studies demonstrated that no one fracture mapping technology can provide a complete picture; rather an integrated diagnostics strategy delivers the most advantageous outcomes. Pressure diagnostics revealed fracture patterns and cluster efficiency that were always a question mark. To improve reliability, floating magnetometer sensors are being developed, which will enable unique stage contribution evaluation and 3D electromagnetic fracture visualization while also scaling up. DNA sequencing and geochemical analysis yield high-resolution vertical layering data, suggesting zone-level production in stacked reservoirs. With over 200 different nanosize tracers, molecular tracers provided effective fluid source monitoring. FDI has been mapped using fiber optics, pressure sensors, and nano-enabled tracers, demonstrating greater cross-stage communication than previously thought. Post-frac diagnostics revealed a substantial association between specific FDI patterns, performance of near-fault stages, and underperforming stages, allowing for targeted completion redesign. The combination of these technologies facilitated better informed, stage-level decision-making, resulting in greater EUR and fewer non-productive clusters. However, cost, operational complexity, and data integration remain key issues, emphasizing the significance of selecting technology that is appropriate for the task at hand.

This study is the initial effort to combine DNA sequencing or geochemical fingerprinting, molecular tracers, fiber optics, sensors and nano-sensors, in addition to an explanation of both emerging technologies: a novel floating magnetometer sensor for stage contribution mapping and 3D fracture geometry mapping for optimal field development. It provides a decision framework for selecting technologies based on reservoir features and operational objectives, hence improving efficiency and sustainability. The magnetometer's stage production, electromagnetic 3D mapping insights, and molecular tracer-driven FDI detection are innovative developments that provide scalable, high-resolution solutions for next-generation hydraulic fracturing and reservoir optimization.

Introduction

Hydraulic fracturing has become a cornerstone in the development of both unconventional and, increasingly, conventional hydrocarbon reservoirs. While significant strides have been made in modeling and operational control, the subsurface remains a dynamic and largely invisible domain. Fracture systems, by nature, are elusive: they propagate in multiple directions, interact with heterogeneities, and evolve in real time.

As engineers and geoscientists, we're often tasked with optimizing completions based on limited or indirect information. This is precisely where fracture mapping technologies become more than just tools; they become enablers of smarter, more informed decisions.

Fracture geometry, including orientation, total and propped length, and fracture height, is controlled by a combination of in-situ stress conditions, rock properties, and operational parameters such as fluid viscosity, pump rate, and proppant concentration. However, the subsurface environment is highly heterogeneous, making it difficult to predict how fractures will propagate. As a result, discrepancies often arise between modeled and actual fracture networks, which directly impacts well's productivity and overall development efficiency.

Historically, single-diagnostic approaches such as microseismic monitoring, *Warpinski, Norm (2009)*, or pressure analysis provided foundational insights into fracture geometry and complexity. But each method comes with trade-offs in scale, resolution, and interpretational ambiguity. The industry's growing awareness of these limitations has driven the development of integrated, multi-scale fracture mapping workflows, leveraging the strengths of each tool while compensating for their blind spots. From distributed fiber-optic sensing to intelligent tracers and AI-augmented modeling, we are witnessing a technological convergence aimed at unlocking a clearer, more dynamic picture of what happens below the surface.

A key issue is identifying the effective fracture geometry, specifically, the portion of the fracture that remains open and propped post-treatment and actually contributes to fluid flow. Despite the use of advanced diagnostics and modeling tools, current fracture mapping technologies (such as microseismic, tracers, or fiber optics) often fall short when used in isolation. These tools may not fully capture the true productive volume unless integrated with geomechanical models, production data, and geological context.

Completion design decisions, including stage count, cluster spacing, and inter-well distance, must account for this uncertainty. Poorly optimized designs can lead to insufficient reservoir contact, fracture interference, and inefficient proppant placement, ultimately reducing recovery and inflating costs.

The implications for field development are significant. Without accurate insight into the true geometry and conductivity of fractures, operators risk underperforming wells, inefficient use of capital, and increased environmental footprint. To address this, a more integrated approach is needed, one that combines improved subsurface characterization, better fracture modeling, and more reliable mapping technologies. Such advances are essential not only to reduce uncertainty, but also to drive more sustainable and economically viable hydraulic fracturing operations.

Fundamental Challenges in Fracture Mapping

Despite years of innovation, mapping fractures in the subsurface remains as much an art as it is a science. The underground world we try to navigate during stimulation treatments is not only out of sight, but also dynamic, heterogeneous, and highly responsive to the smallest changes in stress, pressure, and rock fabric. These complexities, [figure 1](#), pose ongoing challenges that continue to limit our ability to predict, monitor, and optimize hydraulic fracture growth with confidence.

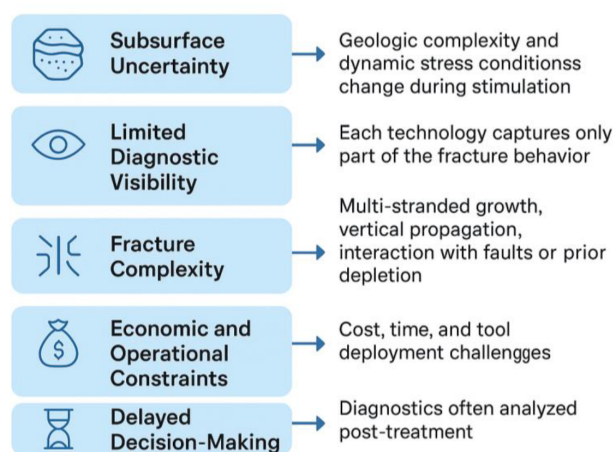


Figure 1—Challenges overview

One of the most fundamental issues is subsurface uncertainty. We often start with a static geologic model, but once high-pressure fluids are introduced, that model becomes a moving target. Natural fractures open or close, stress paths reorient, and new fracture networks form in unpredictable patterns. What we think should happen during pumping doesn't always match what actually occurs underground.

A second major challenge is the limited visibility of most diagnostic tools. Each technology, whether it's microseismic monitoring, pressure analysis, fiber optics, or tracers, offers only a piece of the puzzle. Microseismic shows us fracture initiation and event location, but not necessarily fluid volume or proppant placement. Tracers can reveal stage-by-stage contributions but struggle with spatial accuracy. Even sophisticated tools like DAS/DTS can be constrained by deployment logistics or signal resolution. On their own, these methods often leave us with more questions than answers.

Thirdly, fracture complexity itself is becoming more pronounced, especially in stacked reservoirs and infill development scenarios. Fractures may grow vertically across layers, bifurcate, reorient around faulty intervals, or interact with legacy depletion zones, or fractures driven interactions among wells, [Gupta, I et al \(2021\)](#). The result is often asymmetric, non-planar growth that defies conventional models and complicates interpretation.

Operators also face practical and economic constraints. Advanced diagnostics can be expensive, require specialized hardware, or be limited by wellbore availability. In many cases, data is collected but not fully integrated, either due to time pressures, data silos, or the lack of cross-disciplinary workflows between geoscience, completion, and reservoir teams.

Finally, there's the challenge of timing and decision-making. Most fracture mapping data is either post-job or delayed. Yet, the decisions we need to make, such as when to adjust pumping schedules, alter perforation strategies, or optimize stage spacing, are needed in real time. Bridging that gap between data and decision remains a key hurdle.

In short, while our tools have become smarter, the subsurface has become more complicated, [table 1](#). This tension is exactly why the integration of multi-scale technologies, layered together in a complementary

fashion, is no longer optional. It's becoming essential for operators who want to move beyond guesswork and toward truly data-driven stimulation design.

Table 1—Fundamental Challenges in Hydraulic Fracture Mapping

Challenge	Description	Impact on Fracture Mapping	Typical Limitations in Diagnostics
Subsurface Uncertainty	Geologic complexity and dynamic stress conditions change during stimulation	Difficult to predict fracture geometry and propagation paths	Static models become outdated; real-time adjustments are limited
Limited Diagnostic Visibility	Each technology captures only part of the fracture behavior	Incomplete understanding of geometry, connectivity, and effectiveness	Single datasets can be ambiguous or misleading
Fracture Complexity	Multi-stranded growth, vertical propagation, interaction with faults or prior depletion	Results in asymmetric or unpredictable growth	Conventional models assume planar fractures; reality is often non-planar
Economic and Operational Constraints	Cost, time, and tool deployment challenges	Limits access to high-resolution or permanent diagnostics	Diagnostic choices often driven by budget, not geology
Delayed Decision-Making	Diagnostics often analyzed post-treatment	Missed opportunities for real-time optimization	Lack of live data integration into stimulation workflows

Current Technologies: Capabilities and Limitations

Fracture mapping has come a long way from relying solely on post-frac pressure charts and production logs. Today, a range of specialized tools is available to monitor fracture initiation, growth, and behavior, [Table 2](#). However, no single method can capture the full picture. Each has its own resolution, depth of insight, and operational constraints. In this section, we highlight the most widely used technologies and the role each plays in the fracture diagnostics toolbox. Resolution and Accuracy: Many mapping technologies, such as microseismic monitoring and tiltmeters, struggle with resolution and accuracy. These methods often provide only a rough estimate of fracture's dimensions and orientations, making it challenging to capture the intricate details of the fracture network. The limited resolution can result in an incomplete or distorted representation of the fractures, affecting the accuracy of predictions and models.

Table 2—Summary of Current Fracture Mapping Technologies

Technology	Function	Strengths	Limitations	Ideal Use Case
Microseismic Monitoring	Detects fracture-induced seismic events to map geometry, <i>Maxwell, S.C and Cipolla, C. (2011)</i>	3D fracture growth visualization; complexity & azimuth	No direct measurement of fluid/proppant; costly; requires nearby well	Mapping large-scale fracture networks in high-value developments
Pressure-Based Diagnostics	Interprets treatment pressure to infer near-wellbore behavior, <i>Barree, R. D., et al (2009)</i>	Cost-effective; real-time; universal	Limited far-field insight; model-dependent	Base-level fracture monitoring and treatment calibration
Fiber Optics (DAS/ DTS/DSS)	Uses acoustic, temperature and strain signals along fiber lines, <i>Huckabee, P. (2009) and Jin and Roy (2017)</i>	High-resolution data; detects stage contribution & frac hits	High cost; complex interpretation; installation constraints	Monitoring complex pads; frac hit analysis; stage-by-stage evaluation
Chemical Tracers	Tracks fluid/proppant movement across wells and stages	Reveals stage efficiency; inter-well connectivity; long-term flow paths	Low spatial resolution; delayed data; requires lab analysis	Evaluating drainage connectivity and fracture communication
Tracer Proppant Mapping (Radioactive or not)	Uses proppant coated with gamma-emitting tracers to log near-wellbore proppant placement <i>Holditch, S. A., et al (1993) and McDaniel, R. R., et al (2007)</i>	Direct measurement of near wellbore proppant distribution; highly reliable; proven method	Radiation safety requirements; post-job only; limited to near-wellbore evaluation	Confirming stage/cluster efficiency and diversion effectiveness
Tiltmeters & InSAR	Measures surface deformation to infer subsurface fracture growth, <i>Sercan, G et al (2021)</i>	Large-area monitoring; non-intrusive; ideal for regulatory settings	Low resolution; indirect measurement; best for shallow zones	Broad-scale fracture symmetry and height estimation in sensitive areas

Each of these technologies contributes to a different layer of understanding. However, when used in isolation, they present blind spots that can mislead decision-making. This is why the next wave of innovation is focused not just on improving individual diagnostics, but on integrating them into cohesive, multi-scale workflows.

Emerging Technologies and Integration Trends

The complexity of modern reservoirs, especially in stacked, depleted, or parent-child well environments, demands more than just traditional diagnostics. A new wave of technologies is reshaping how we observe, interpret, and respond to fracture behavior. These tools go beyond geometry, offering insights into reservoir depletion, fluid origin, and even unseen pathways. In this section, we explore some of the most promising innovations that are redefining fracture mapping.

- **Microseismic Depletion Delineation:** While microseismic monitoring has long been used to track fracture growth, recent advancements now allow us to extract depletion zone boundaries from seismic quiet zones, *Dohmen, Ted et al (2014)*. In essence, the absence of events becomes as valuable as their presence, [figure 02](#). This method identifies areas where pressure has already been drawn down, helping operators optimize well spacing, avoid fracture interference, and better understand parent-child interactions. *Why it matters: Depletion-induced stress changes alter fracture direction and effectiveness. Mapping these zones with microseismic provides a proactive lens into fracture containment and reactivation risks.*
- **Time-Lapse Geochemical Fingerprinting:** This technique involves analyzing the chemical composition of produced fluids over time to infer changes in the fracture network. By analyzing stable isotopes, rare elements, and chemical signature, operators can gain insight into the connectivity and effectiveness of fractures, and dynamic production height, [figure 03](#). This method is particularly useful for understanding the long-term behavior of fractures and optimizing production strategies. *Why it matters: Operators can now detect whether production fluids originate from near-field fractures or extended drainage, making it possible to assess stimulation effectiveness beyond early flowback data.*
- **Electromagnetic Fluids Fracture Mapping (EMFM):** Among the more unconventional yet powerful diagnostic techniques, EMFM leverages time-lapse resistivity imaging to detect the movement of saline stimulation fluids during and after hydraulic fracturing. This surface-based method transmits low-frequency electromagnetic waves into the subsurface, where the contrast between electrically resistive rock and conductive frac fluids (especially brine or slickwater) creates a detectable signal. By tracking how these signals evolve, operators can monitor fluid migration, fracture containment, and interwell communication, without deploying downhole tools, [figure 4](#). Providers have demonstrated its effectiveness in the Permian Basin, offering a non-intrusive way to map fluid pathways over time and enhance understanding of fracture behavior in complex geology. *Why it matters: It enables operators to monitor fluid placement and containment in real time—even in wells without fiber or downhole sensors, by simply using surface-based measurements that detect changes in subsurface conductivity caused by salinity.*
- **Acoustic High-Definition Imaging:** enables operators to perform volumetric perforation erosion analysis using 3D measurements of entry and exit hole geometries. Unlike traditional methods that rely solely on exit hole diameters (2D), this technology captures the entire perforation tunnel volume, producing detailed models of proppant erosion pathways throughout the casing, [figure 5](#). The method generates empirical, high-fidelity diagnostics, yet at a significantly lower cost and without requiring real-time infrastructure. It allows us to see inside the wellbore post-stimulation with millimeter precision, even in the absence of live data feeds. By quantifying tunnel growth volumetrically, engineers can now assess treatment uniformity, proppant distribution efficiency,

and stage performance with a level of clarity previously reserved for fiber-equipped wells. *Why it matters: This technology provides high-resolution, cost-effective insight into perforation and proppant behavior, comparable to fiber optics, without the need for real-time infrastructure, enabling smarter, data-driven completion decisions.*

- DNA Production Mapping:** It uses microbial DNA from cuttings while drilling and compares with produced fluids as a natural tracer to map fracture height and produced fluid flow from hydraulic fracturing, offering a high-resolution view of subsurface dynamics, [figure 6](#). By sequencing DNA and comparing it to a reservoir's genetic "fingerprint" collected from cores or cuttings, pinpoints which reservoir layers are producing, constrains fracture heights and lengths, and detects issues like frac hits or water breakthrough, especially in complex fields. This approach complements traditional methods like microseismic monitoring by integrating with pressure and production data, leveraging high-throughput sequencing and bioinformatics to refine completion designs and optimize recovery. Challenges include interpreting complex DNA data and initial costs, but automation and growing databases are making it more accessible. *Why it matters: it empowers operators to maximize production efficiency and minimize environmental risks with unprecedented precision.*
- Nanosized tracers:** It revolutionizes fracture mapping by injecting smart nanoparticles into frac fluids or proppants to track fluid and proppant flow in real time, providing precise insights into fracture geometry, stage-level production, and inter-well communication like fracture driven interaction, [Shokanov, T et al \(2020, 2021 and 2023\)](#). These tracers, over 220 unique, deployed in multistage fracs and analyzed via surface sampling and sub-atomic measurements, reveal drainage volumes and diverter efficiency, as demonstrated in our well identifying interaction among wells and impact from faults for completion optimization, [figure 7, Melo, R. C. B et al \(2024\)](#). Integrated with AI-driven analytics, this approach offers a cost-effective alternative to traditional tracers and even DAS, enhancing completion designs in complex reservoirs. *Why it matters: it drives smarter, more efficient fracturing operations, maximizing production while minimizing costs and risks, moreover only one tracer per stage would be used for oil and water.*
- Machine Learning and Data Integration:** The integration of machine learning algorithms with mapping technologies has revolutionized the interpretation of fracture data. Machine learning models can analyze large datasets from various sources, such as microseismic, tiltmeters, and fiber optics, to provide more accurate and comprehensive fracture maps. This approach reduces uncertainties and enhances the predictive capabilities of fracture models.

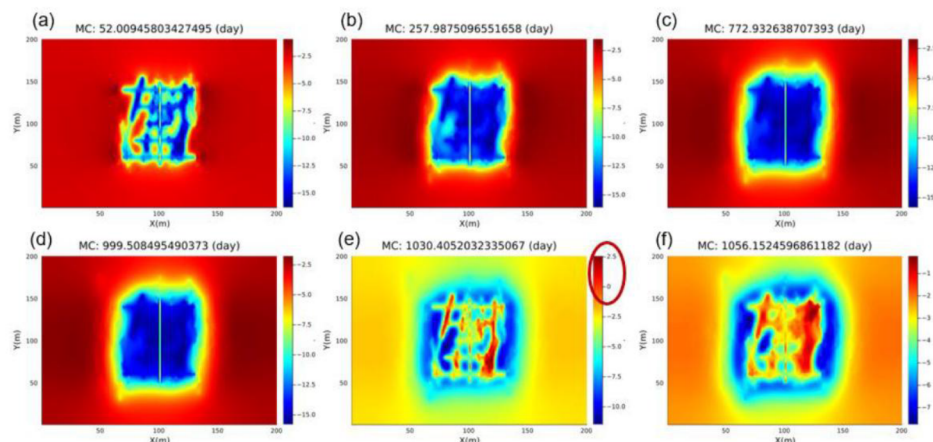


Figure 2—The principal maximum horizontal stress computed during 1000 days of depletion, followed by 100 days reinjection with liquid fluid. The units are MPa, and the initial in-situ stress was 88.435 MPa, [Shabelansky, Andrey et al 2022](#)

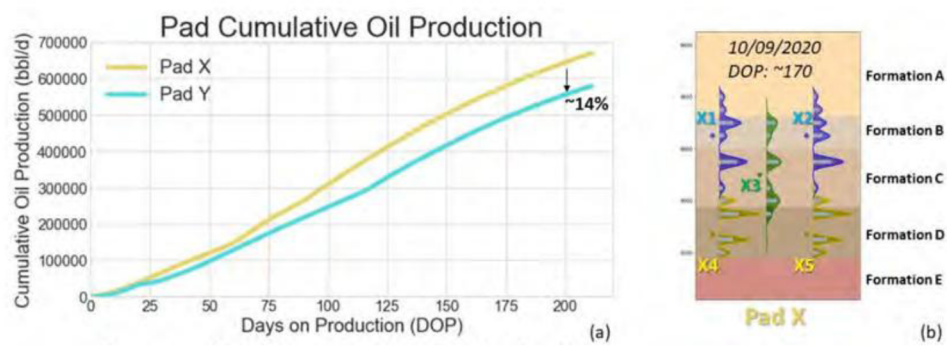


Figure 3—Pad cumulative oil production and markers of all the 5 wells in Pad X, [Ge, Shuangyu et al \(2022\)](#)

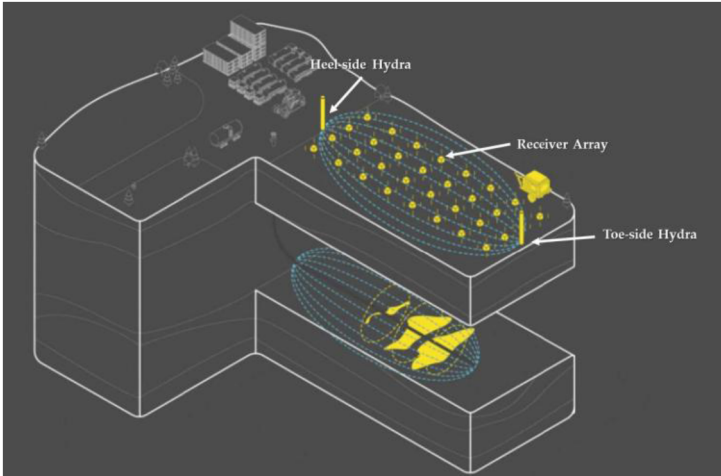


Figure 4—Schematic of the surface EM array. Two hydrams on the surface connected by a transmitter line and a receiver array on both sides designed to cover the desired lateral extent in the subsurface. Injection of conductive frac fluid perturbs the baseline EM field in the subsurface around the wellbore, [Reynolds, Amanda et al \(2022\)](#)

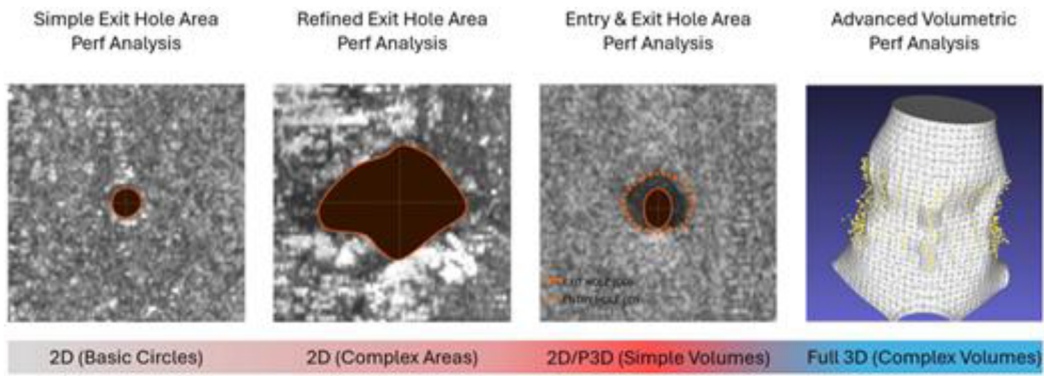


Figure 5—Acoustic High-Definition Imaging, [Jones, Matt et al \(2025\)](#)

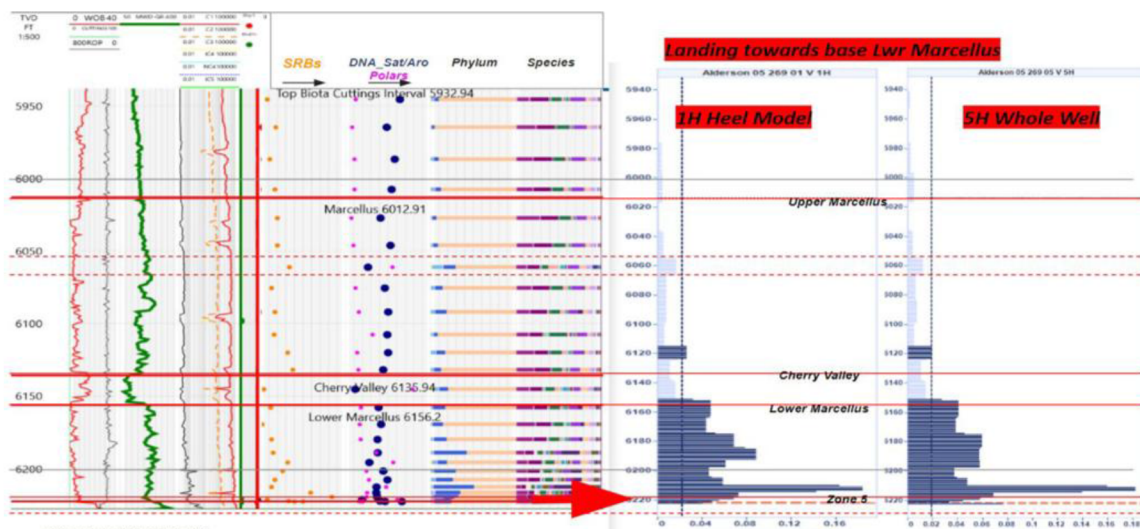


Figure 6—DNA production mapping in Marcellus, dynamic height production to evaluate Cherry Valley Limestone pinch out of the created fracture, report courtesy from Biota.

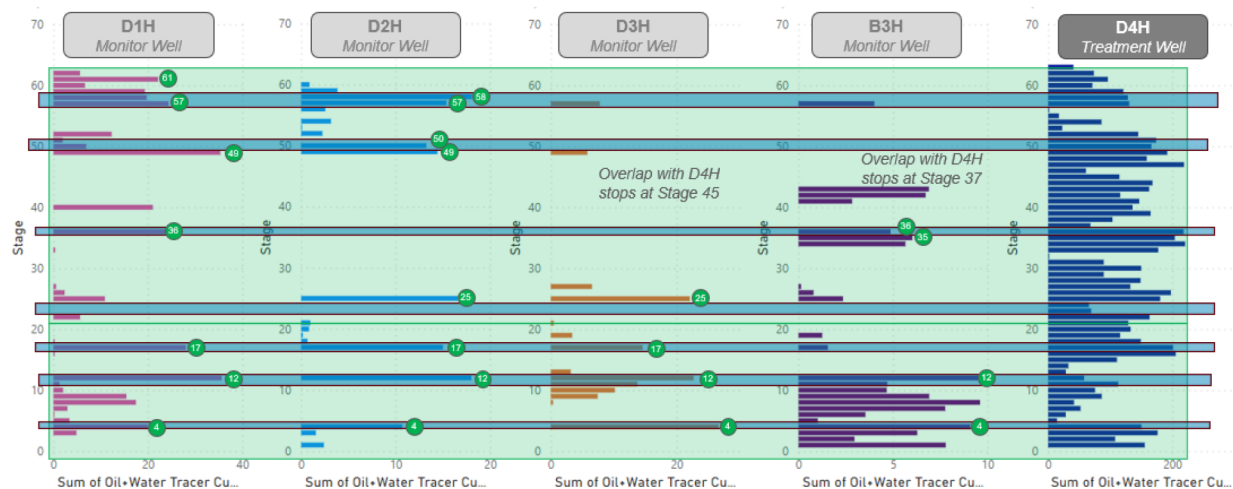


Figure 7—Cumulative tracer recovery in monitoring and offset wells, Melo, R. C. B et al (2024).

These advancements in mapping technologies are helping to address the limitations of traditional methods, providing more accurate and reliable data for optimizing hydraulic fracturing operations. By leveraging these innovations, operators can enhance the efficiency, effectiveness, and sustainability of their field development efforts.

The Future Outlook: Next-Generation Mapping Paradigms

The future of fracture mapping is poised to leap forward with next-generation paradigms that blend cutting-edge sensor technologies, electromagnetic proppant mapping innovations, and AI-driven analytics, promising a level of precision and efficiency that feels like something from fiction science. Imagine tiny, floating sensors, drifting through frac fluids to map subsurface dynamics in real time, or electromagnetic proppants provide illumination for tracing fracture networks and proppant placement.

Floating Magnetometer

One exciting ongoing development is the technology outlined in patent US11629590B2, Melo RCB et al (2023), which describes a method and system for estimating flowback or reservoir fluid production

rates in oil and gas wellbores using magnetic irregularity elements, such as casing collars, and floating sensor elements. This time released sensors, dragged by the fluid flow within the wellbore's casing, detect magnetic perturbations caused by ferromagnetic casing collars or other magnetic elements arranged along the wellbore's path, [figure 8](#). By measuring the time, it takes for sensors to pass between known locations of these magnetic elements, the method calculates fluid velocity and, consequently, the flow rate of individual well segments or the entire well, leveraging data from multiple wells for enhanced accuracy. This non-invasive approach improves upon traditional methods like chemical, temperature or pressure sensors by offering precise, real-time flow rate estimates, optimizing completion designs and production efficiency in complex reservoirs.

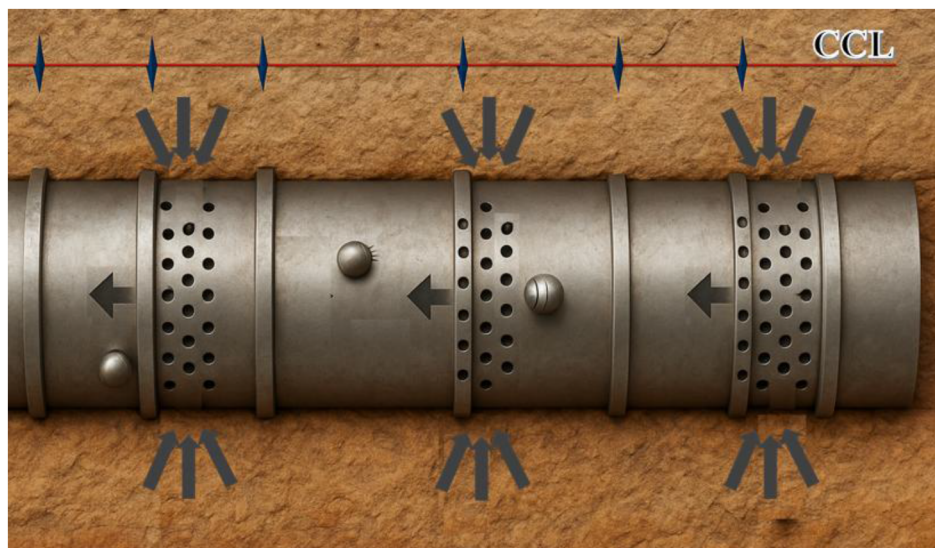


Figure 8—CCL floating sensors flowing inside of casing, capturing each cluster/stage production contribution.

Electromagnetic Proppant Mapping

Meanwhile, the Advanced Energy Consortium (AEC) is pushing the envelope with electromagnetic proppant mapping, [LaBrecque et al \(2016\)](#), and the pressure transient analyses and 3D pore-elastic modeling to assess fracture dilation and stress, [Haddad, M and Ahmadian, M, \(2023\)](#), using conductive or magnetic proppants to create detectable signals that reveal fracture special propped extent and connectivity with stunning detail, [figure 9](#). Validated in a South Texas field test, these proppants act like glowing markers, helping operators visualize fracture patterns and fluid flow, optimizing well designs while reducing environmental impact. The technique is a small-scale field experiment conducted to develop electromagnetic techniques for tracking proppant placement and effects in hydraulic fractures. The study tested proppants like Loresco coke breeze and steel shot in shallow, sub-horizontal fractures at 1.5 m depth than a second trial at 70m, using electrical impedance tomography (EIT) with dense electrode arrays to image fracture geometries. Results showed that geophysical inversions accurately reflect horizontal-shaped fractures (3-5 m in extent, ~6 mm thick). The tests were validated by core samples (deeper test) and excavation (Shallower), with conductivity responses enhanced by coke breeze and phase changes notable with steel shot, though inversion smoothing underestimated conductance. The second experiment described by [Haddad, M and Ahmadian, M, \(2023\)](#) used pressure transient analysis (PTA) and 3D poro-elastic modeling to understand how repeated water injections affect fracture dilation and bottomhole pressure (BHP). They analyzed data from six injections and developed a finite-element model that incorporates both propped and unpropped fracture zones. Their simulations showed that accounting for these zones and the cumulative effects of previous injections significantly improves the prediction of BHP. They also highlight the importance of fracture permeability, formation properties, and the initially open fracture area in

matching field data. Ultimately, the work supports the development of a multiphysics model that integrates electromagnetic (EM) signals and streaming potential (SP) to better interpret surface-recorded electric fields during hydraulic fracturing operations.

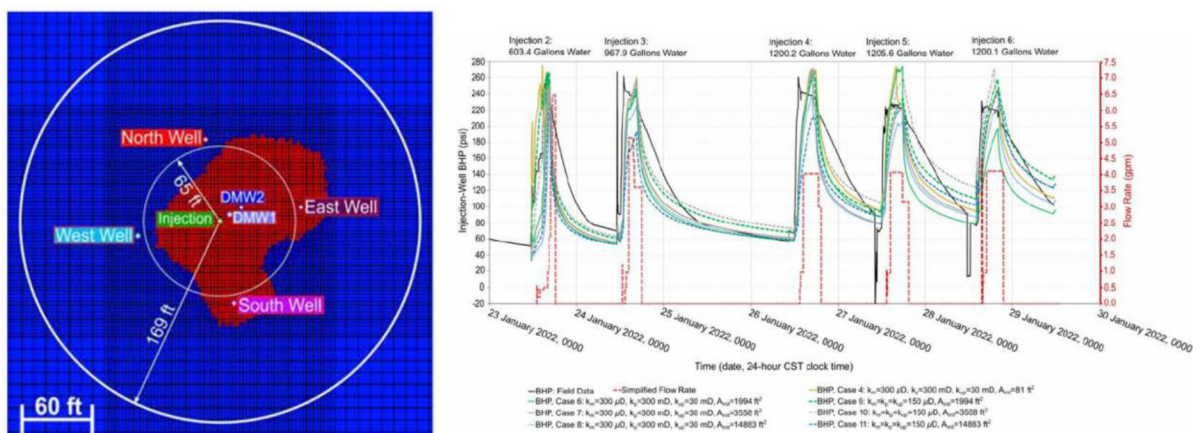


Figure 9—Right -Schematic of the propped horizontal hydraulic fracture at the DFPS (red area), the injection and monitoring wells, the propped-fracture-area equivalent circle (65 ft radius), and the total-fracture-area equivalent circle (169 ft)- Haddad et al. (2023). Left - Comparison of the injection-well BHP from the field data and the simulations to evaluate the explicit effect of the formation permeability on the injection-well BHP at various fracture areas, Haddad et al. (2023).

These advancements are like upgrading from a paper map to a 3D, interactive "GPS" for the subsurface, promising smarter, more efficient mapping. AI integration can be set to accelerate these innovations by enhancing how we process and interpret data from advanced sensors and mapping technologies. Machine learning algorithms, like those inspired by physics-informed neural networks, can sift through complex magnetic and electromagnetic datasets to predict fracture behavior, optimize completion designs, and even forecast production outcomes with remarkable accuracy. By combining floating magnetometer data, electromagnetic proppant signals, others mapping technologies mentioned in this paper, and AI, operators will be able to create dynamic, real-time models of fracture networks, adapting on the fly to reservoir conditions. This synergy not only will boost recovery rates but also minimizes costly missteps, like frac hits or water breakthroughs. The future of fracture mapping lies in this fusion of smart sensors and intelligent analytics, turning the subsurface into a transparent, "manageable" system. This matters because it paves the way for smarter, safer, and more sustainable energy extraction, balancing economic gains with environmental responsibility.

Integration of technologies for effective frac mapping. Cases Histories

Case 1: Integrated Hydraulic Fracture Mapping Using Microseismic, Fiber Optic, Tracers, Pressure Decline, and Clustering Analytics

Note: Due to confidentiality constraints, the specific field and geographical location associated with this case study cannot be disclosed.

This case study presents a multi-scale hydraulic fracture monitoring campaign that integrated surface and downhole microseismic, high-density geophone arrays, machine learning-enhanced event detection, pressure decline analysis, fiber optics (DAS, DSS and DTS) and production tracer diagnostics. The objective was to characterize fracture behavior and optimize the completion strategy across multiple horizontal wells by monitoring all the 3 wells (A, B and C), for this it was attempted several different completion strategies in Well A, such as cluster number, spacing, diverters, proppants, fluids, etc.

A total of 22,895 seismic events were confirmed through a robust workflow involving enhanced stacking, manual pick refinement, and machine learning catalog enhancement. The monitoring array, composed of

virtual 3C stations arranged in hexagonal super-stations, achieved magnitude completeness thresholds down to approximately -0.8 , with location accuracy medians of 145 m (X), 171 m (Y), and 288 m (Z). Moment tensor inversion revealed a predominance of thrust faulting mechanisms, consistent with a stress regime where $SH_{max} > S_v$, and strike orientations aligned NW–SE (Figure 10). Those sub-seismic features, generate a fracture corridor once the induced fracture hit such feature, this resulted in re-evaluate the volume of proppant and well spacing. The derived PGV–Mw relationship enabled magnitude estimation for smaller events, extending interpretive capabilities beyond the limits of spectral fitting (Figure 11).

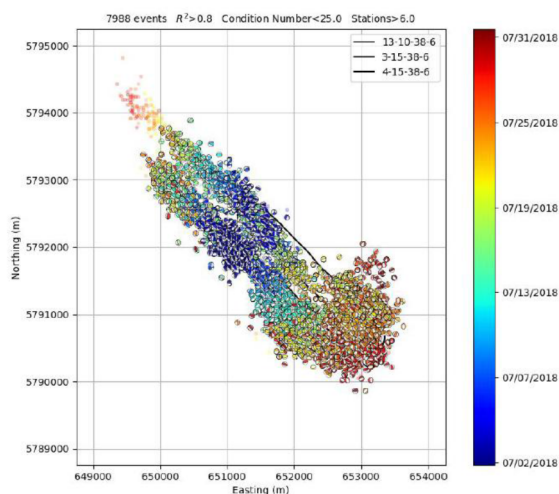


Figure 10—Moment Tensor "Beach Balls". Spatial distribution and orientation of mechanisms

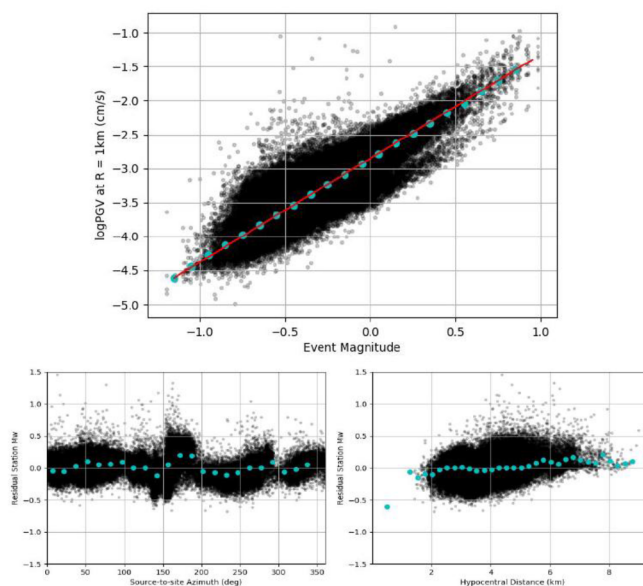


Figure 11—PGV–Mw Relationship: Regression model and residual analysis

Clustering analysis using the FAST algorithm identified waveform-similar event families, improving location precision and revealing consistent mechanisms within clusters (Figure 12). Fiber optic data revealed multiple frac hits, though only a subset resulted in true hydraulic communication, as validated by tracer returns.

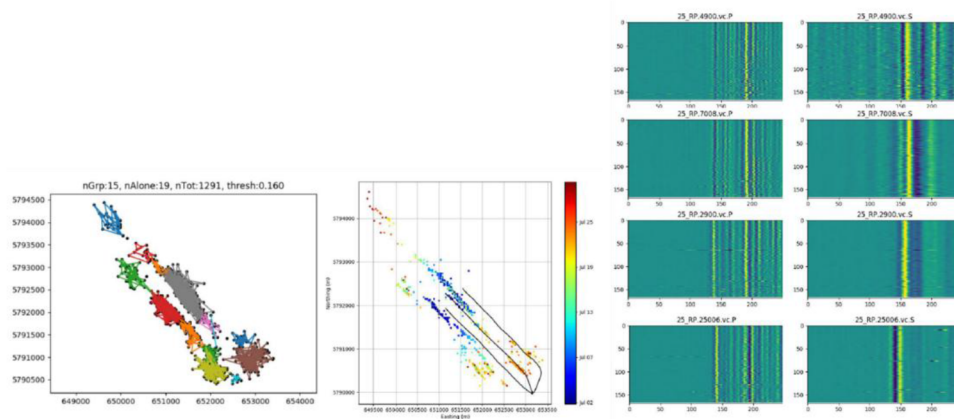


Figure 12—FAST Clustering Results. Clustered event locations (left) and waveform stacks (right)

The production tracers offered a unique lens into both the quality and connectivity of stimulated intervals. By assigning specific tracers to different stages, we could move beyond guesswork and qualitatively pinpoint which zones were contributing most effectively to production. The tracer returns didn't just reveal productive sweet spots within a well; they also illuminated inter-well communication on the pad, highlighting cases where stimulation in one well "A", in middle, led to measurable production interference, or even unplanned fluid migration, in adjacent laterals, wells B and C (Figure 13). This ability to "see" both the best intervals and cross-well dynamics in real time gave us actionable insights, not only for optimizing completion designs but also for safeguarding production efficiency across the entire pad.

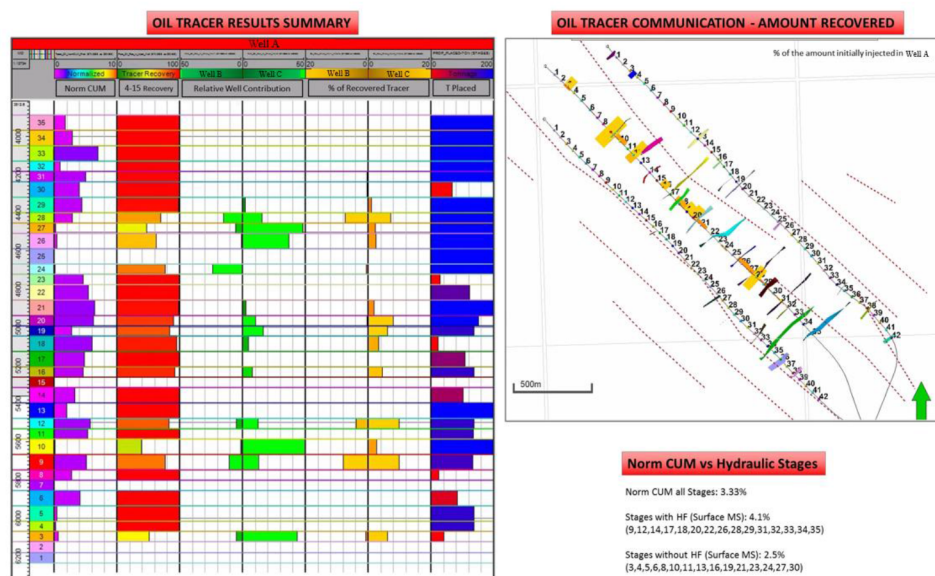


Figure 13—Production tracer: left the frequency of return and right the X-flow.

Fiber optic data revealed multiple frac hits, though only a subset resulted in true hydraulic communication, as validated by tracer returns. Beyond merely identifying fracture-driven interactions with fiber optic measurements, after the initial finding of DAS cross-well responses by Webster *et al.* (2013), Jin and Roy (2017) discovered that the low-frequency DAS signal (LF-DAS) can be used as a hybrid DSS to monitor strain perturbation induced by hydraulic fracture propagation at offset monitor wells, Jin, G. (2022). The integration of DSS and DAS provided a powerful lens into strain distortion in observation well generated by FDIs from the DSS and the spatial distribution of proppant within each cluster per stage with DAS. Fiber optic data, through DSS, DAS and DTS, revealed not only the presence of frac hits,

figure 14, but enabled differentiation between mere pressure communication and true hydraulic connectivity, validated by the arrival and return of chemical tracers. This approach allowed for the identification of best completion strategy, corresponding to robust fracture openness and efficient proppant distribution, figure 15. Conversely, clusters with muted DSS responses and poor tracer recovery highlighted areas of limited proppant access, guiding future design improvements. By pairing fiber optic diagnostics with proppant distribution insights, the study achieved a nuanced, cluster-level understanding of stimulation effectiveness, enabling operators not only to optimize completion designs but also to enhance reservoir connectivity and overall possible production performance.

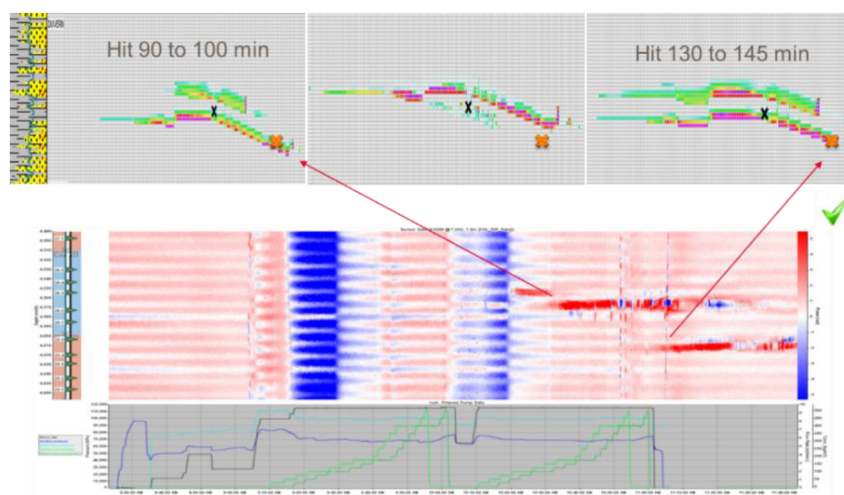


Figure 14—Fracture Driven Interaction evaluation with DSS, supporting the post frac pressure match and fracture geometry propagation

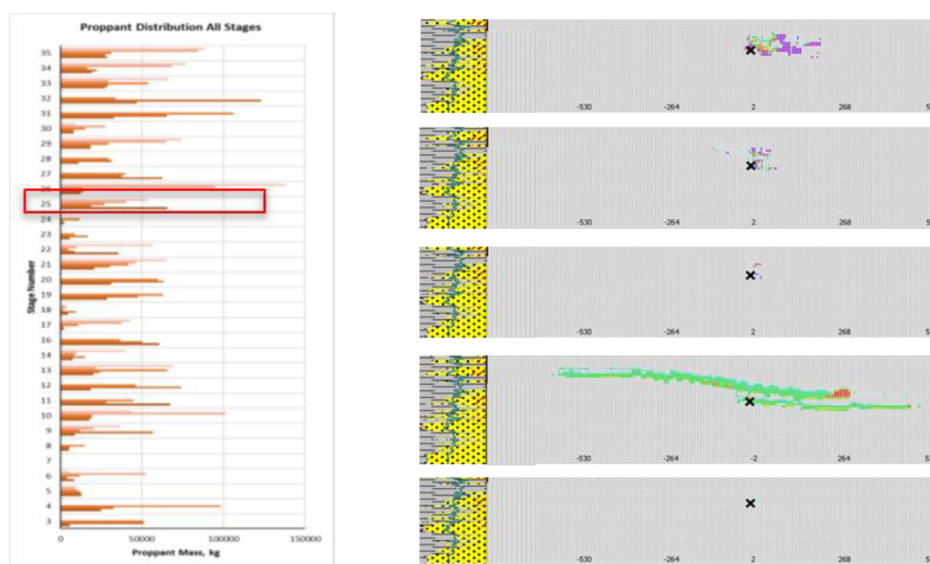


Figure 15—Cluster Efficiency evaluation supporting the clustering efficiency for post frac pressure match

Pressure's decline matching confirmed fracture geometry and leak-off behavior, with adjustments to anisotropy and leak-off coefficients improving model fidelity (Figure 16).

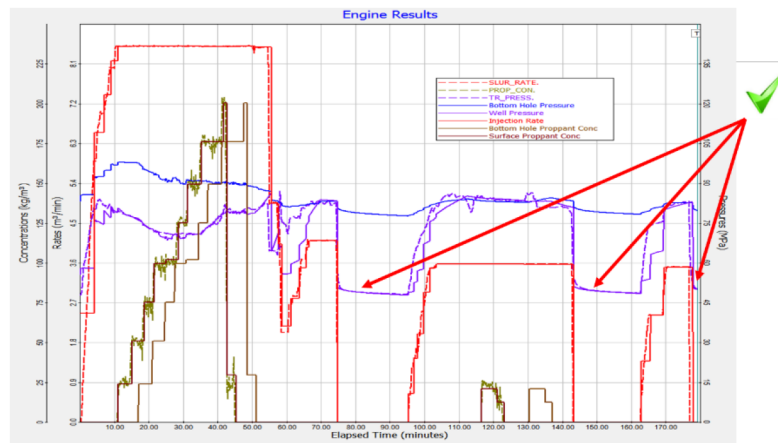


Figure 16—Pressure Decline Matching. Example of matched pressure curves and geometry

Completion optimization studies demonstrated that increasing cluster spacing and pumping rates improved proppant distribution, while the use of diverters enhanced stage coverage and reduced screen-out risks. Recommendations included initiating stimulation from the central well, avoiding stage overlap, and maintaining diverter use to improve cluster activation. The integration of these technologies provided a high-resolution, multi-dimensional view of fracture behavior, enabling more effective stimulation strategies and improved reservoir performance.

Case 2: Multi-Scale Characterization of Hydraulic Fractures Using DAS, DTS, Microseismic, and Direct Sampling in the Eagle Ford Shale

This case study presents a comprehensive hydraulic fracture characterization pilot conducted by ConocoPhillips in the Eagle Ford shale, [Rateman et al., \(2017\)](#). The project integrated microseismic monitoring, distributed acoustic and temperature sensing (DAS/DTS), pressure gauges, radioactive tracers, and direct sampling via deviated wells to evaluate the geometry, complexity, and proppant distribution within the Stimulated Rock Volume (SRV) adjacent to a horizontal producer.

The pilot included five deviated observation wells and one vertical monitor well surrounding a central producer. Over 7,700 ft of SRV were sampled using core and image logs, revealing a dense network of steeply dipping, NE/SW-oriented hydraulic fractures. Many fractures formed swarms and exhibited branching at bedding interfaces ([Figures 14](#)). Fracture intensity was found to decrease more rapidly with height than with lateral distance, suggesting a laterally broad SRV geometry ([Figure 15](#)).

Microseismic monitoring captured over 26 stages, with events clustering near the wellbore and extending up to 1,500 ft laterally. DAS and DTS data provided high-resolution insights into fracture initiation, fluid distribution, and stress shadowing effects ([Figures 6–10](#)). DAS strain signals correlated with fracture opening and closure, while DTS confirmed fluid entry across all clusters. However, plug leakage was observed in 10 of 13 stages, affecting stage isolation and fluid efficiency.

Proppant distribution was assessed via radioactive tracers and cuttings analysis. While proppant was detected in 76% of samples from the closest well (S3), only 5% of samples from the furthest well (ST02) contained proppant. Embedded proppant was confirmed in only two hydraulic fractures ([Figure 16](#)), indicating sparse emplacement and potential challenges in maintaining fracture conductivity.

Correlation analysis between microseismic event density and hydraulic fracture density showed limited statistical agreement, with moderate correlation only for fractures dipping less than 70° ([Figure 17](#)). Similarly, DAS-derived fluid volumes showed negligible correlation with fracture density, suggesting that microseismic and DAS data alone may not reliably predict fracture intensity or permeability.

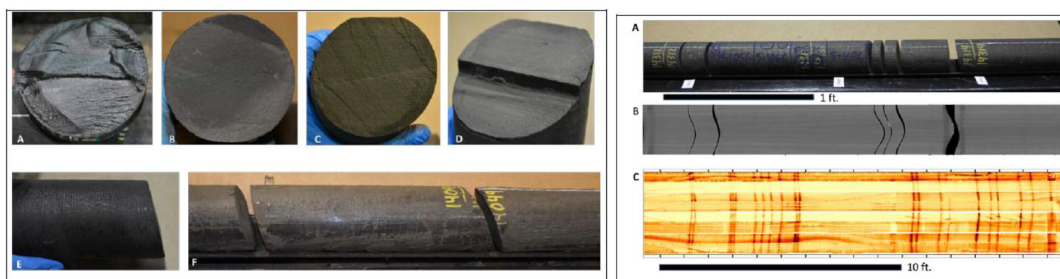


Figure 17—Left: A-Drilling induced fractures evidenced in cores. B to F-Hydraulic fractures. Right: A-Dipping fractures in core. B-Same section of CT image as an unwrapped circumferential image. C-18 ft. section of FMI-HD image log containing several fractures. Taken from [Rateman et al., \(2017\)](#).

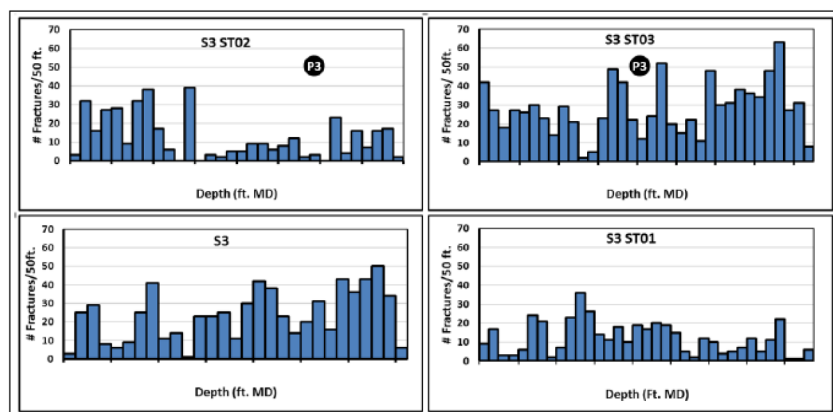


Figure 18—Hydraulic fracturing pattern. "P3" denotes the point where the well trajectory intersects above the P3 producing well. Taken from [Rateman et al., \(2017\)](#).

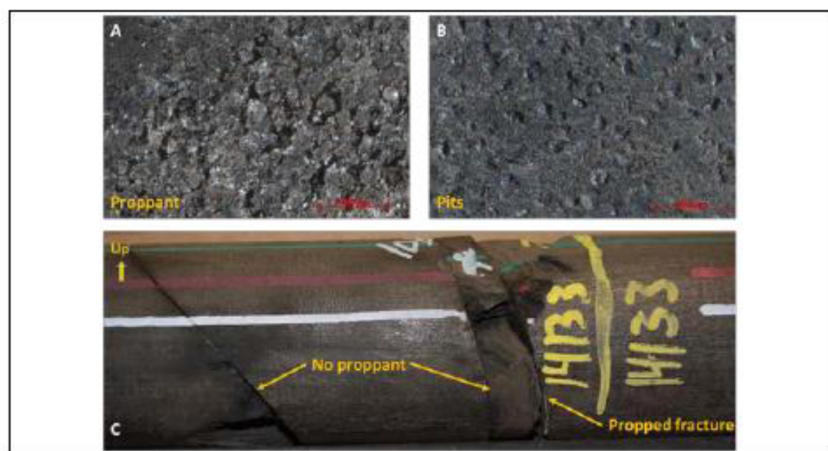


Figure 19—Proppant embedment A) Sand grains on surface, B) Embedment pits C) Fracture in core. Taken from [Rateman et al., \(2017\)](#).

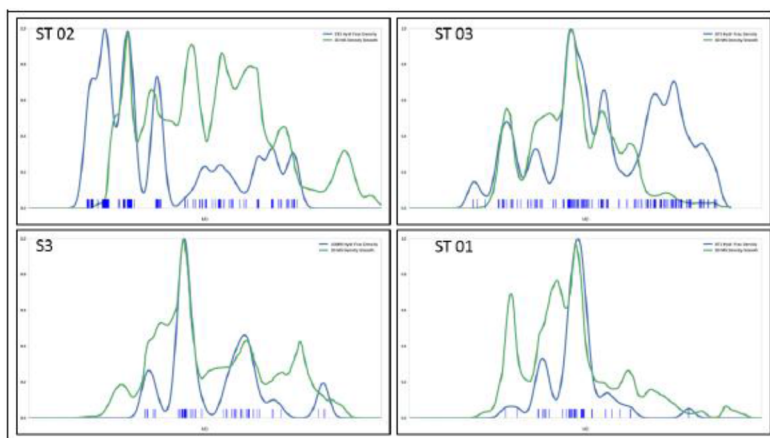


Figure 20—Fracture distribution along the well paths. Green curve is Gaussian Kernel of microseismic density. Blue curve is Gaussian Kernel of hydraulic fracture density and actual fracture location along each wellbore is shown by blue vertical lines at the base of each plot. Taken from [Rateman et al., \(2017\)](#).

The two case studies presented in this paper exemplify the transformative impact of integrating multi-scale monitoring technologies in hydraulic fracture mapping. Each project, one in a tight reservoir using microseismic, pressure, and tracer data, and the other in the Eagle Ford shale employing direct SRV sampling, DAS/DTS, and microseismic, demonstrates how combining diverse datasets yields a more complete and actionable understanding of fracture behavior.

In the first case, the synergy between surface and downhole microseismic, machine learning-enhanced event detection, and tracer diagnostics enabled high-resolution mapping of fracture propagation and connectivity. The integration of moment tensor inversion and clustering analytics further refined interpretations of fracture mechanisms and stress orientations, while pressure decline analysis validated fracture geometry and fluid efficiency. This multi-disciplinary approach led to optimized completion strategies and improved reservoir performance.

The Eagle Ford pilot extended this paradigm by incorporating direct physical sampling of the SRV through strategically placed deviated wells. The combination of core analysis, image logs, DAS/DTS, and radioactive tracers provided unprecedented insight into fracture density, geometry, and proppant distribution. The findings revealed a complex, laterally extensive fracture network with limited proppant emplacement and highlighted the limitations of relying solely on microseismic or DAS data to infer fracture intensity or permeability.

Together, these case histories reinforce a central message: the integration of complementary monitoring technologies enables smarter, more efficient fracture mapping. By capturing fracture behavior across spatial and temporal scales, operators can move beyond simplified models and toward data-driven designs that enhance stimulation effectiveness, reduce uncertainty, and maximize reservoir drainage. These advancements mark a significant step forward in the evolution of hydraulic fracturing diagnostics and underscore the value of multi-scale integration in achieving optimal reservoir performance.

Conclusions

As hydraulic fracturing evolves to meet the demands of increasingly complex reservoirs, fracture mapping must do more than reveal geometry, it must guide engineering with clarity and confidence. Today's diagnostic landscape is richer than ever, blending traditional microseismic, proppant markers (radioactives) and pressure analysis with high-resolution fiber optics, sophisticated tracers (either injected from surface or original markers from reservoir), electromagnetic sensing, and machine learning. Each method contributes a unique layer of insight. Yet, despite this technological progress, true integration remains a frontier not fully crossed. Most operators still work with fragmented datasets, constrained by logistical, financial, or

real-time limitations. We have the tools, but we often lack the connected workflows and adaptive decision systems to turn data into a timely, actionable strategy.

To fully unlock the performance of our wells and reservoirs, the next step must be integration, not just innovation. It's no longer about having the latest sensor or tracer, it's about creating workflows that let geoscientists, engineers, and data scientists collaborate in real time. The way forward lies in breaking the silos between diagnostics and design. This means embracing hybrid workflows that fuse of the mapping technologies to give as many dimensions as we need, 3D frac geometry, effective and dynamic fracture geometry, clustering/stages efficiency, cross communication, and data science, not just during analysis and execution but during production life. Future success depends not only on what we can see underground, but on how fast and intelligently we respond to what we see. By investing in multi-scale, real-time, and interoperable mapping systems, and embedding them into our completion and reservoir workflows, we can move from post-job understanding to true fracture steering. The industry is not short on innovation; it is short on integration. Closing that gap will define the next decade of unconventional success.

What We Have Today:

- A broad diagnostic toolbox: microseismic, DAS/DTS/DSS, EM sensing, tracers, and pressure analysis
- High-resolution post-job imaging (e.g., Acoustic High-Definition Imaging, Nano particle tracers)
- Machine learning models for post-treatment analysis
- Advanced multi-tracers enabling zone-specific mapping
- Time-lapse geochemical and DNA fingerprinting for fluid tracking
- Valuable data from offset wells and legacy activity

What We're Still Missing:

- Seamless integration of diagnostics into live stimulation decision-making
- Cross-domain workflows between geoscience, completions, and data teams
- Affordable, scalable diagnostic solutions for all well types — not just premium cases
- Real-time dashboards and adaptive algorithms that close the loop during pumping
- A culture shift: from observing the past to shaping the present

We must transition from passively monitoring fractures to actively steering them, using live data to guide treatments, update models, and shape completion decisions on the fly. This shift demands a mindset change: from diagnostics as insurance to diagnostics as an investment; from post-job learning to in-job adaptation. When we combine technical excellence with operational agility, we don't just understand our fractures better, we make them work better.

By focusing on integration, simplification, and real-time collaboration, the industry can move from data-rich to decision-rich. This is the next step in the evolution of fracture diagnostics, and it's not just about better visibility underground, but better outcomes above it.

Acknowledgement

We would like to express our gratitude to the Society of Petroleum Engineers (SPE) for providing the opportunity to publish this paper in their prestigious conference. We would also like to acknowledge the invaluable contributions of our colleagues who assisted with the data collection, analysis, and interpretation, without which this study would not have been possible.

Furthermore, we would like to extend our appreciation to the management of Repsol for authorizing the publication of this paper, and for their continued support of our efforts. We also acknowledge the contributions of the reviewers and editors for their valuable feedback and suggestions, which have greatly improved the quality of this paper.

Finally, we extend our thanks to all those who have contributed to this study in ways that may not be mentioned here. Your support and encouragement have been essential to our success.

Nomenclature

3D	Three Dimensional
ABS	Acrylonitrile Butadiene Styrene
AEC	Advanced Energy Consortium
AI	Artificial Intelligence
BHP	Bottomhole Pressure
DAS	Distributed Acoustic Sensing
DNA	Deoxyribonucleic Acid
DSS	Distributed Strain Sensing
DTS	Distributed Temperature Sensing
EIT	Electrical Impedance Tomography
EM	Electromagnetic
EUR	Estimated Ultimate Recovery
FAST	Features from Accelerated Segment Test
FDI	Frac-Driven Interactions
FMI-HD	Formation Microimager, High Definition
LF-DAS	Low-Frequency DAS
Mw	Moment Magnitude
NE-SW	Northeast - Southwest
NW-SE	Northwest - Southeast
PGV	Peak Ground Velocity
PTA	Pressure Transient Analysis
RFS	Rayleigh Frequency Shift
S/A	Sociedad Anónima (Public Limited Company, as used in Spain)
SP	Streaming Potential
SPE	Society of Petroleum Engineers
SRV	Stimulated Rock Volume

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